RRG Sector Rotation

Technical Appendix: Data, Methods, and Supplementary Results

Companion to the five-page English report 7 November 2026

This appendix uses the revised historical ETF pool throughout. Q denotes pure first-quadrant RRG; C denotes diffusion + RRG; M denotes sector momentum; E denotes equal-weight ETFs. Unless a table changes the setting, the sample is 28 December 2018–31 March 2026, with 220/60/20 trading-day signal windows, a maximum of six sectors, and 3 bp one-way fees. Percentages describe returns or risk; pp denotes percentage points. Tables and English vector figures are generated from the archived full-precision results. The conventions and comparison families match those of the main report.

A Data conventions and execution audit

A.1 Historical classification and product eligibility

Historical membership is matched to its applicable interval. Missing classifications are excluded from the affected calculation. Free-float value is computed as reported free-float shares times the unadjusted stock close, with the source share-unit conversion. Its equivalence with Wind's free-float measure remains a data-definition issue. Stock-level diffusion uses unadjusted price changes. Sector signals use industry indices.

Stock-day coverage below is classified stock-days divided by observed stock-days; market-value coverage is their free-float value divided by the total available free-float value. The 2026 row ends on 31 March. Additional queries returned the existing 6,750 membership intervals and added zero intervals.

Table A1: Historical CITIC sector-membership coverage

Year	Stock-days	Classified days	Day coverage	Value coverage
2017	743,941	389,554	52.36%	37.64%
2018	817,583	457,242	55.93%	35.55%
2019	885,332	530,937	59.97%	41.89%
2020	946,205	914,685	96.67%	98.66%
2021	1,059,106	1,021,817	96.48%	99.13%
2022	1,146,505	1,118,625	97.57%	99.42%
2023	1,209,700	1,189,567	98.34%	99.59%
2024	1,233,444	1,227,173	99.49%	99.86%
2025	1,248,134	1,242,166	99.52%	99.88%
2026	289,895	288,473	99.51%	99.85%

The product universe uses industry/theme labels in the current index directory. Matching exchange-traded codes to the corresponding off-exchange fund records repairs 248 metadata fields. Across the sample, 718 funds meet type/date eligibility; 54 are currently delisted and 35 previously enter an effective historical candidate pool. Current labels and incomplete historical tracking-index identities introduce classification uncertainty. At each formation, observed quotations, listing dates, holdings availability, overseas-holdings filtering, and same-index turnover selection determine eligibility. Twenty-day average turnover uses available history for more recent listings.

A.2 Holdings and delisted-fund treatment

Fund exposures use holdings records announced by formation and historical sector membership. Partial disclosures retain original NAV weights. The fixed implementation deduplicates revised announcements using the latest announcement record, which can affect historical holdings availability. These conventions reduce certain information-timing risks while leaving incomplete disclosure and revision-history limitations.

Suspended or delisted holdings are valued at the last available adjusted price until modeled settlement. At the first monthly formation on or after delisting, the latest NAV already announced by formation, multiplied by the historical adjustment factor, determines estimated proceeds on the next trading day; fees apply. Plans cover 34 candidates. Main-setting E has 25 estimated settlements; Q, C and M each have zero. In the parameter grid, M at 260/6 and 260/8 has one each. Actual liquidation dates and cash receipts may differ.

A.3 Research-report reconstruction and audit scope

The source report uses Wind and Gildata. This study uses an authorized compatible interface and a reconstructed industry/theme ETF pool. The following difference describes the aggregate effect of data, universe and implementation conventions.

Table A2: Composite ETF strategy: published and reconstructed performance

Metric	Published	Reconstructed
Annualized return	21.51%	18.95%
Annualized volatility	24.93%	27.02%
Maximum drawdown	−34.98%	−38.40%
Sharpe	0.800	0.778
Calmar	0.620	0.493

Archived validation records report successful cash, fee, ending-value, next-open and post-settlement checks for all 43 cost/grid runs. Four main daily NAV paths reconcile with the prior revised-pool run; the three 220/6 grid paths reconcile with their corresponding main paths. These are accounting and implementation checks conditional on the inputs. Continuous units, proportional fees and adjusted opening prices are used. Lot sizes, minimum commissions and capital-specific market impact require further implementation work.

B Signal definitions, optimizer, and performance measures

Let $P_{i,t}$ denote the sector price index, B_t the daily-rebalanced equal-weight 28-sector benchmark NAV, and MA_{20} a trailing 20-trading-day simple average. The adopted fixed code calculates

$$\begin{aligned} B_t &= B_{t-1}\left(1+\frac{1}{n_t}\sum_{i\in I_t}\left(\frac{P_{i,t}}{P_{i,t-1}}-1\right)\right), \\ RS_{i,t} &= 100P_{i,t}/B_t, \\ R_{i,t} &= \mathrm{MA}_{20}\left(100RS_{i,t}/RS_{i,t-L}\right), \\ M_{i,t} &= \mathrm{MA}_{20}\left(100R_{i,t}/R_{i,t-60}\right). \end{aligned}$$

Here I_t is the set with available returns and $n_t = |I_t|$. Main $L = 220$. Q selects up to $N = 6$ sectors with $R > 100$ and $M > 100$, ranked by $\sqrt{(R-100)^2+(M-100)^2}$. Coordinate values exactly equal to 100 are left outside the strict quadrant inequalities. M ranks sectors by $P_{i,t}/P_{i,t-L} - 1$.

For sector i, diffusion is the 20-day smoothed fraction of available free-float value in constituent stocks with positive L-day price changes:

$$D_{i,t} = \mathrm{MA}_{20}\left[\frac{\sum_{j\in i} v_{j,t}\,\mathbf{1}(p_{j,t}/p_{j,t-L}-1>0)}{\sum_{j\in i} v_{j,t}}\right].$$

C ranks diffusion, takes the top N sectors, and retains those in the first or second quadrant. Its selection stops at that filter, so actual counts vary. Valid denominators with zero rising constituents give zero diffusion. The repair resides in the data adapter.

Let $E_{f,i}$ be visible industry exposure per ETF NAV, s_i the binary selected-sector indicator, and $k = \sum_i s_i$. The three active schemes share:

$$\begin{aligned} \max_{w} \quad & \sum_f w_f \sum_i E_{f,i}s_i \\ \text{subject to} \quad & \sum_f w_f = 1, \quad 0 \le w_f \le 1, \\ & \sum_f w_f E_{f,i} \le 1/k \quad \text{for each selected sector } i. \end{aligned}$$

This caps selected-sector exposure while allowing unequal ETF weights, exposure to other sectors and concentrated ETF holdings. The main Q portfolio's time-average largest actual ETF weight is 42.72%; for C it is 50.50%. E rebalances the available exposure-matrix fund pool to equal ETF weights.

With $T = 1756$, annualized return is $(V_T/V_0)^{252/T} - 1$. Volatility is the daily-return standard deviation times $\sqrt{252}$. Sharpe is daily mean return divided by its standard deviation times $\sqrt{252}$, with zero risk-free rate. Maximum drawdown is the minimum of $V_t/\max_{u\le t} V_u - 1$. Calmar divides annualized return by absolute maximum drawdown. Two-way turnover adds purchases and sales divided by prior NAV and annualizes by $252/T$.

Table B1: Supplementary main-setting implementation statistics

Strategy	Calmar	Two-way turnover	Mean largest ETF	Ending CNY m
Q	0.550	14.97	42.72%	46.36
C	0.493	15.21	50.50%	33.51
M	0.473	12.56	41.95%	39.58
E	0.261	1.77	0.97%	19.57

C Time variation and return concentration

Table C1: Calendar-period net returns

Period	Q	C	M	E
2019	57.45%	27.27%	55.82%	31.10%
2020	93.54%	69.09%	67.06%	34.30%
2021	26.52%	27.60%	20.66%	7.60%
2022	-16.47%	-1.67%	-9.90%	-20.58%
2023	-16.55%	-21.07%	-21.34%	-7.31%
2024	19.22%	10.93%	16.88%	10.61%
2025	37.59%	30.33%	41.80%	29.02%
2026 Q1	5.15%	8.77%	7.30%	-1.66%

The 2018 observation is 28 December only, before first execution; all four returns are zero. The 2026 Q1 observation is an unannualized three-month return.

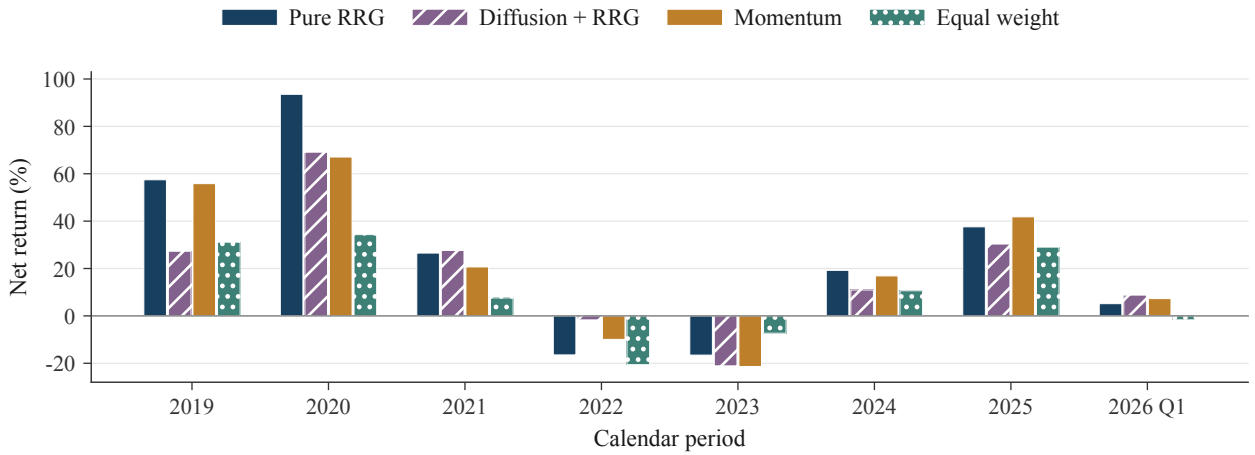


Figure C1: Calendar-period net returns after 3 bp one-way costs. The final observation covers January–March 2026 and is presented as an unannualized quarterly return.

Table C2: Rolling 12-month descriptive statistics

Strategy	Windows	Ranked first	Positive	Median	Worst	Best
Q	76	29	50	20.22%	-23.81%	93.54%
C	76	18	51	21.56%	-22.77%	72.01%
M	76	13	52	14.30%	-25.07%	71.10%
E	76	16	48	9.41%	-24.52%	53.33%

Adjacent rolling windows overlap. Q exceeds M, C and E in 47, 53 and 55 windows, respectively, based on the archived rolling monthly series.

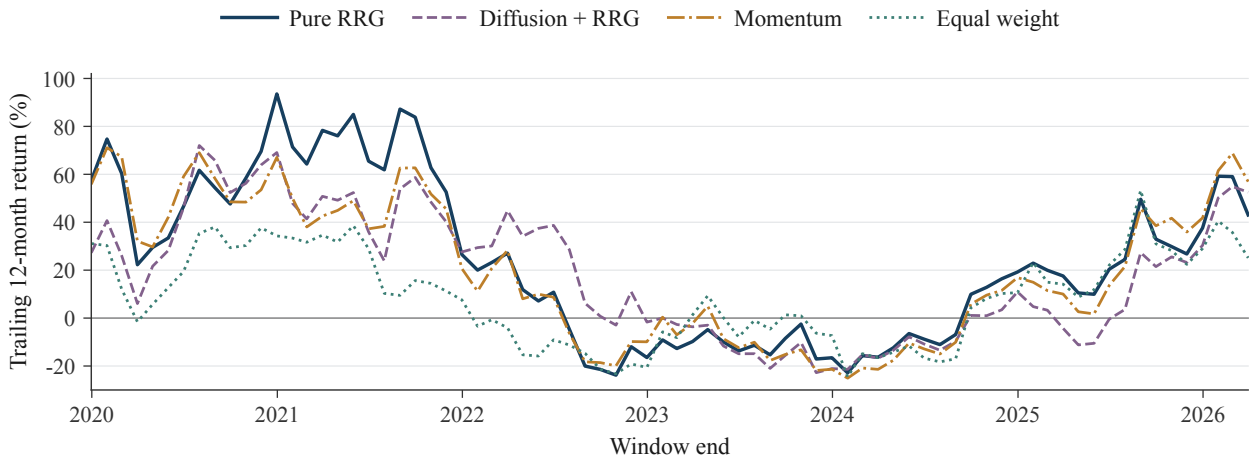


Figure C2: Trailing 12-month net returns. Each series comprises 76 monthly observations; adjacent windows share eleven months. The horizontal reference line denotes zero return.

Table C3: Month-concentration diagnostic

Strategy	Positive/negative	Top-3 log growth	Best-1 flat CAGR	Best-3 flat CAGR
Q	46/41	34.92%	20.97%	15.40%
C	48/39	46.37%	15.51%	9.75%
M	47/40	35.11%	18.92%	13.67%
E	49/38	73.31%	6.90%	2.61%

Top-three growth share divides the sum of the best three monthly log returns by full-period log growth. Flat-month scenarios set each strategy's own best one or three monthly returns to zero while retaining sample length. They diagnose retrospective concentration.

Table C4: Best and worst return months

Strategy	Best month	Return	Worst month	Return
Q	2020-12	23.01%	2020-03	-12.65%
C	2021-08	22.73%	2020-03	-12.97%
M	2025-08	18.33%	2020-03	-13.37%
E	2024-09	22.95%	2024-01	-12.31%

D Cost, parameter, and statistical robustness

D.1 Cost and parameter sensitivity

Each cost case replays the same target rules with uniform one-way fees of 3, 5, 10 or 20 bp and zero additional slippage. The parameter grid changes $L \in \{180, 220, 260\}$ and $N \in \{4, 6, 8\}$. The 60-day momentum and 20-day smoothing windows remain fixed. C changes both diffusion and relative-strength lookback with L. E retains its fixed main return. All results use the same observed historical sample.

Table D1: All 16 transaction-cost cases

Strategy	bp	Ann. return	Max. DD	Volatility	Sharpe	Calmar
Q	3	24.62%	-44.76%	26.36%	0.967	0.550
C	3	18.95%	-38.40%	27.02%	0.778	0.493
M	3	21.83%	-46.19%	25.92%	0.892	0.473
E	3	10.12%	-38.78%	19.68%	0.588	0.261
Q	5	24.25%	-45.11%	26.35%	0.956	0.538
C	5	18.59%	-38.90%	27.01%	0.767	0.478
M	5	21.52%	-46.57%	25.91%	0.883	0.462
E	5	10.08%	-38.81%	19.68%	0.587	0.260
Q	10	23.32%	-45.98%	26.34%	0.928	0.507
C	10	17.69%	-40.11%	26.99%	0.739	0.441
M	10	20.77%	-47.49%	25.90%	0.859	0.437
E	10	9.99%	-38.89%	19.68%	0.582	0.257
Q	20	21.49%	-47.69%	26.33%	0.871	0.451
C	20	15.91%	-42.78%	26.96%	0.683	0.372
M	20	19.27%	-49.28%	25.87%	0.811	0.391
E	20	9.80%	-39.10%	19.67%	0.574	0.251

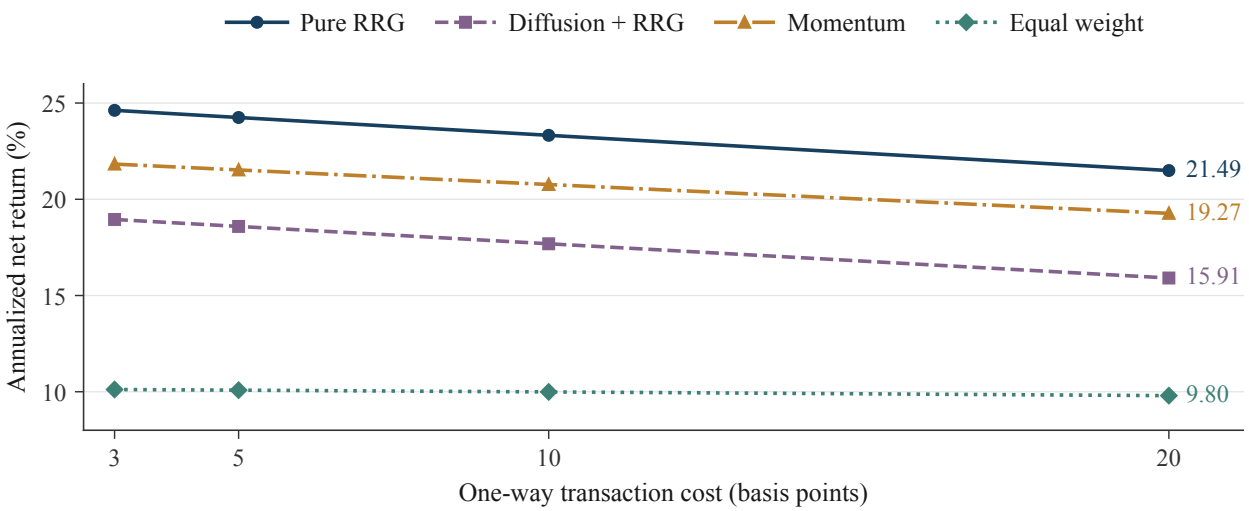


Figure D1: Transaction-cost sensitivity. Target portfolios are replayed at each uniform one-way cost. Labels at 20 bp report the resulting annualized net returns (%).

Table D2: All 27 active-strategy parameter cases

Strategy	L	N	Ann. return	Max. DD	Volatility	Sharpe	Calmar
Q	180	4	20.14%	-47.18%	27.23%	0.811	0.427
C	180	4	16.91%	-43.05%	27.59%	0.705	0.393
M	180	4	19.59%	-55.43%	28.29%	0.774	0.353
Q	180	6	19.40%	-44.01%	25.54%	0.822	0.441
C	180	6	12.07%	-35.27%	26.25%	0.566	0.342
M	180	6	16.76%	-46.96%	25.30%	0.739	0.357
Q	180	8	16.27%	-39.06%	25.01%	0.728	0.417
C	180	8	14.04%	-41.86%	25.56%	0.642	0.335
M	180	8	13.58%	-48.57%	24.23%	0.647	0.280
Q	220	4	17.42%	-48.83%	27.76%	0.718	0.357
C	220	4	23.98%	-44.83%	29.91%	0.869	0.535
M	220	4	18.81%	-48.94%	26.86%	0.776	0.384
Q	220	6	24.62%	-44.76%	26.36%	0.967	0.550
C	220	6	18.95%	-38.40%	27.02%	0.778	0.493
M	220	6	21.83%	-46.19%	25.92%	0.892	0.473
Q	220	8	23.95%	-37.57%	25.65%	0.966	0.637
C	220	8	20.16%	-42.93%	26.27%	0.831	0.470
M	220	8	19.94%	-44.49%	25.11%	0.850	0.448
Q	260	4	12.73%	-57.89%	28.25%	0.566	0.220
C	260	4	15.08%	-58.74%	28.93%	0.631	0.257
M	260	4	15.50%	-54.11%	27.67%	0.660	0.287
Q	260	6	21.14%	-54.01%	27.19%	0.842	0.391
C	260	6	16.62%	-48.01%	27.37%	0.699	0.346
M	260	6	19.38%	-48.48%	26.54%	0.801	0.400
Q	260	8	19.16%	-55.91%	26.86%	0.787	0.343
C	260	8	13.71%	-56.65%	27.54%	0.605	0.242
M	260	8	19.80%	-46.51%	25.79%	0.830	0.426

Matched parameters mean the same lookback and nominal sector cap; realized counts may differ. Q exceeds matched M in six cases and matched C in seven. It exceeds fixed 220/6 M in two cases. Q is above E in all nine cases. These comparisons answer different allocation questions.

D.2 Paired mean-return inference

Tests use the 87 non-overlapping complete months from January 2019 through March 2026. For paired return differences, HAC inference uses lag three as the main choice, small-sample correction and a two-sided t approximation; lags one and six are sensitivity checks. Confidence intervals are individual 95% intervals. Holm adjustment is applied separately to six primary, 24 cost and 54 grid comparisons. Every adjusted p-value in those three families exceeds 0.05. These families cover the specified current tests; historical research choices add further selection uncertainty.

Table D3: Six primary HAC(3) paired comparisons

Pair	Mean gap pp	95% CI pp	Raw p	Holm p
Q–C	0.395	[-0.404, 1.194]	0.3284	0.9852
Q–M	0.198	[-0.342, 0.739]	0.4681	0.9852
Q–E	1.076	[0.199, 1.953]	0.0168	0.1010
C–M	-0.197	[-0.896, 0.503]	0.5773	0.9852
C–E	0.681	[-0.345, 1.706]	0.1904	0.7615
M–E	0.878	[-0.062, 1.817]	0.0667	0.3335

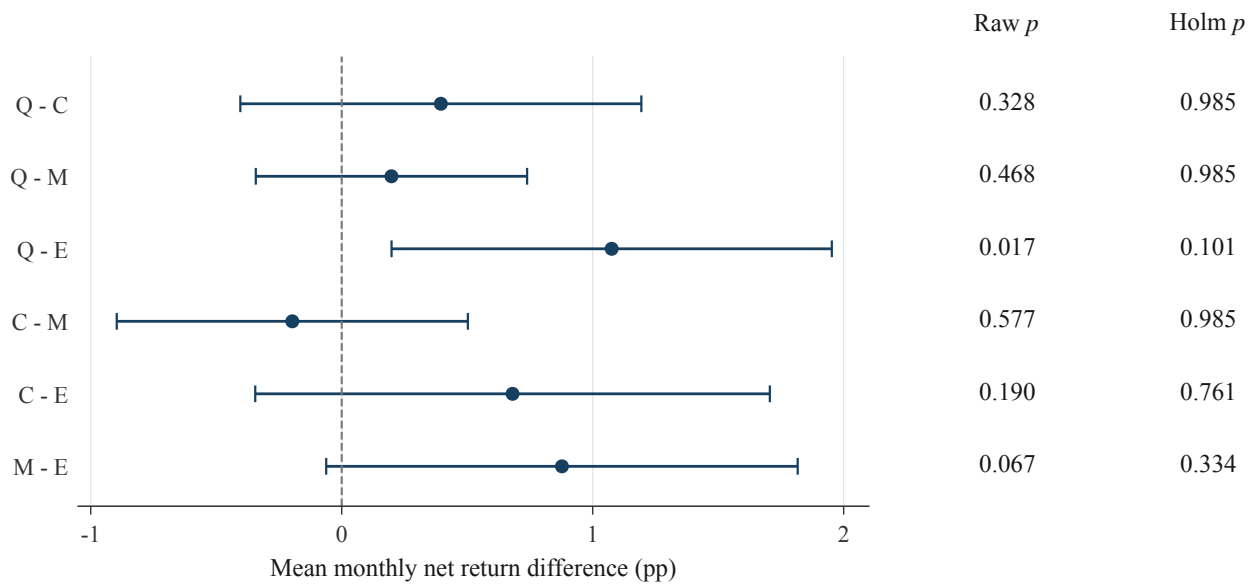


Figure D2: Paired mean monthly net-return differences. Points denote estimates and bars individual 95% HAC(3) confidence intervals. Raw p-values and Holm-adjusted p-values refer to the six primary comparisons. Units are percentage points.

Table D4: HAC lag sensitivity: all 18 cases

Lag	Pair	Mean gap pp	95% CI pp	Raw *p*
1	Q–C	0.395	[-0.322, 1.113]	0.2768
3	Q–C	0.395	[-0.404, 1.194]	0.3284
6	Q–C	0.395	[-0.482, 1.272]	0.3729
1	Q–M	0.198	[-0.353, 0.749]	0.4763
3	Q–M	0.198	[-0.342, 0.739]	0.4681
6	Q–M	0.198	[-0.354, 0.750]	0.4772
1	Q–E	1.076	[0.106, 2.045]	0.0301
3	Q–E	1.076	[0.199, 1.953]	0.0168
6	Q–E	1.076	[0.234, 1.917]	0.0128
1	C–M	-0.197	[-0.932, 0.539]	0.5960
3	C–M	-0.197	[-0.896, 0.503]	0.5773
6	C–M	-0.197	[-0.945, 0.551]	0.6021
1	C–E	0.681	[-0.388, 1.750]	0.2090
3	C–E	0.681	[-0.345, 1.706]	0.1904
6	C–E	0.681	[-0.371, 1.732]	0.2016
1	M–E	0.878	[-0.138, 1.893]	0.0895
3	M–E	0.878	[-0.062, 1.817]	0.0667
6	M–E	0.878	[0.012, 1.743]	0.0471

Table D5: Cost-family paired comparisons: all 24 cases

bp	Pair	Mean gap pp	95% CI pp	Raw *p*	Holm *p*
3	Q–C	0.395	[-0.404, 1.194]	0.3284	1.0000
3	Q–M	0.198	[-0.342, 0.739]	0.4681	1.0000
3	Q–E	1.076	[0.199, 1.953]	0.0168	0.4038
3	C–M	-0.197	[-0.896, 0.503]	0.5773	1.0000
3	C–E	0.681	[-0.345, 1.706]	0.1904	1.0000
3	M–E	0.878	[-0.062, 1.817]	0.0667	1.0000

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Table D5 continued

bp	Pair	Mean gap pp	95% CI pp	Raw p	Holm p
5	Q–C	0.395	[-0.404, 1.195]	0.3282	1.0000
5	Q–M	0.194	[-0.346, 0.734]	0.4768	1.0000
5	Q–E	1.054	[0.177, 1.931]	0.0191	0.4396
5	C–M	-0.201	[-0.901, 0.498]	0.5686	1.0000
5	C–E	0.658	[-0.367, 1.684]	0.2054	1.0000
5	M–E	0.860	[-0.080, 1.800]	0.0724	1.0000
10	Q–C	0.396	[-0.404, 1.197]	0.3278	1.0000
10	Q–M	0.183	[-0.354, 0.721]	0.4992	1.0000
10	Q–E	0.999	[0.122, 1.876]	0.0260	0.5726
10	C–M	-0.213	[-0.914, 0.488]	0.5473	1.0000
10	C–E	0.603	[-0.424, 1.629]	0.2463	1.0000
10	M–E	0.816	[-0.125, 1.756]	0.0883	1.0000
20	Q–C	0.398	[-0.405, 1.202]	0.3272	1.0000
20	Q–M	0.162	[-0.371, 0.695]	0.5464	1.0000
20	Q–E	0.891	[0.015, 1.768]	0.0464	0.9738
20	C–M	-0.236	[-0.938, 0.467]	0.5061	1.0000
20	C–E	0.493	[-0.535, 1.521]	0.3433	1.0000
20	M–E	0.729	[-0.213, 1.671]	0.1278	1.0000

Table D6: Parameter-family paired comparisons: all 54 cases

L	N	Pair	Mean gap pp	95% CI pp	Raw p	Holm p
180	4	Q–C	0.179	[-0.683, 1.041]	0.6812	1.0000
180	4	Q–M	0.044	[-0.741, 0.828]	0.9122	1.0000
180	4	C–M	-0.135	[-1.197, 0.927]	0.8010	1.0000
180	4	Q–E	0.804	[-0.171, 1.780]	0.1050	1.0000
180	4	C–E	0.625	[-0.386, 1.637]	0.2225	1.0000
180	4	M–E	0.760	[-0.473, 1.994]	0.2238	1.0000
180	6	Q–C	0.512	[-0.237, 1.261]	0.1777	1.0000
180	6	Q–M	0.226	[-0.605, 1.057]	0.5901	1.0000
180	6	C–M	-0.286	[-0.968, 0.396]	0.4066	1.0000
180	6	Q–E	0.725	[-0.127, 1.577]	0.0942	1.0000
180	6	C–E	0.213	[-0.686, 1.112]	0.6389	1.0000
180	6	M–E	0.499	[-0.437, 1.435]	0.2919	1.0000
180	8	Q–C	0.147	[-0.383, 0.677]	0.5820	1.0000
180	8	Q–M	0.238	[-0.452, 0.929]	0.4943	1.0000
180	8	C–M	0.091	[-0.481, 0.663]	0.7524	1.0000
180	8	Q–E	0.495	[-0.269, 1.258]	0.2016	1.0000
180	8	C–E	0.347	[-0.478, 1.173]	0.4055	1.0000
180	8	M–E	0.256	[-0.524, 1.036]	0.5156	1.0000
220	4	Q–C	-0.479	[-1.180, 0.221]	0.1774	1.0000
220	4	Q–M	-0.113	[-0.762, 0.536]	0.7300	1.0000
220	4	C–M	0.366	[-0.472, 1.204]	0.3873	1.0000
220	4	Q–E	0.611	[-0.266, 1.489]	0.1695	1.0000
220	4	C–E	1.091	[-0.086, 2.267]	0.0687	1.0000
220	4	M–E	0.724	[-0.459, 1.908]	0.2269	1.0000
220	6	Q–C	0.395	[-0.404, 1.194]	0.3284	1.0000
220	6	Q–M	0.198	[-0.342, 0.739]	0.4681	1.0000
220	6	C–M	-0.197	[-0.896, 0.503]	0.5773	1.0000
220	6	Q–E	1.076	[0.199, 1.953]	0.0168	0.8918
220	6	C–E	0.681	[-0.345, 1.706]	0.1904	1.0000
220	6	M–E	0.878	[-0.062, 1.817]	0.0667	1.0000

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Table D6 continued

L	N	Pair	Mean gap pp	95% CI pp	Raw p	Holm p
220	8	Q–C	0.261	[-0.214, 0.737]	0.2774	1.0000
220	8	Q–M	0.276	[-0.193, 0.744]	0.2456	1.0000
220	8	C–M	0.014	[-0.566, 0.595]	0.9612	1.0000
220	8	Q–E	1.014	[0.285, 1.743]	0.0069	0.3741
220	8	C–E	0.753	[-0.150, 1.655]	0.1010	1.0000
220	8	M–E	0.738	[-0.004, 1.480]	0.0511	1.0000
260	4	Q–C	-0.163	[-0.919, 0.594]	0.6706	1.0000
260	4	Q–M	-0.178	[-1.102, 0.746]	0.7026	1.0000
260	4	C–M	-0.016	[-0.839, 0.808]	0.9702	1.0000
260	4	Q–E	0.325	[-0.551, 1.202]	0.4628	1.0000
260	4	C–E	0.488	[-0.729, 1.704]	0.4276	1.0000
260	4	M–E	0.503	[-0.761, 1.768]	0.4311	1.0000
260	6	Q–C	0.336	[-0.353, 1.024]	0.3356	1.0000
260	6	Q–M	0.152	[-0.613, 0.917]	0.6937	1.0000
260	6	C–M	-0.184	[-0.817, 0.449]	0.5657	1.0000
260	6	Q–E	0.895	[-0.069, 1.859]	0.0685	1.0000
260	6	C–E	0.559	[-0.381, 1.500]	0.2406	1.0000
260	6	M–E	0.743	[-0.214, 1.700]	0.1266	1.0000
260	8	Q–C	0.373	[-0.155, 0.901]	0.1636	1.0000
260	8	Q–M	-0.001	[-0.668, 0.667]	0.9986	1.0000
260	8	C–M	-0.374	[-1.015, 0.267]	0.2498	1.0000
260	8	Q–E	0.757	[-0.212, 1.727]	0.1241	1.0000
260	8	C–E	0.384	[-0.507, 1.276]	0.3935	1.0000
260	8	M–E	0.758	[-0.142, 1.658]	0.0978	1.0000

D.3 Paired circular block bootstrap

Monthly return pairs are resampled together in circular blocks of three or six months, with 10,000 repetitions and random seed 20261107. Centered two-sided bootstrap p-values refer to the mean monthly difference. The table separately reports the annualized compounded-return gap and its percentile interval, annualized with the original $252/1756$ exponent. This preserves the distinction between the two estimands.

Table D7: All 12 bootstrap cases

Pair	Block	Mean gap pp	Centered p	CAGR gap pp	CAGR 95% CI pp
Q–C	3	0.395	0.3083	5.67	[-5.331, 17.626]
Q–C	6	0.395	0.3506	5.67	[-6.350, 18.500]
Q–M	3	0.198	0.4591	2.79	[-4.846, 10.761]
Q–M	6	0.198	0.4803	2.79	[-4.850, 11.573]
Q–E	3	1.076	0.0203	14.50	[2.543, 29.729]
Q–E	6	1.076	0.0099	14.50	[3.139, 27.925]
C–M	3	-0.197	0.5620	-2.88	[-13.186, 6.771]
C–M	6	-0.197	0.5953	-2.88	[-14.022, 7.766]
C–E	3	0.681	0.1964	8.83	[-4.881, 24.950]
C–E	6	0.681	0.1934	8.83	[-4.863, 23.727]
M–E	3	0.878	0.0733	11.71	[-0.765, 27.464]
M–E	6	0.878	0.0490	11.71	[0.020, 25.390]

D.4 Investable-benchmark regressions

For Q and C separately, monthly returns are regressed on a constant, M returns and E returns. Intercept inference uses HAC(3). This measures increment relative to two investable portfolio benchmarks; its interpretation remains conditional on their exposures.

Table D8: Benchmark regressions

Strategy	Intercept pp	95% CI pp	Raw p	Holm p	R^2
Q	0.304	[-0.160, 0.768]	0.1956	0.3913	0.879
C	0.066	[-0.638, 0.770]	0.8520	0.8520	0.759

For Q, the M and E coefficients are 0.823 and 0.230; for C, they are 0.783 and 0.139. Sector, size and other factor risks remain part of the unexplained attribution question.

E Fixed ten-month moving-average risk-control comparison

The common market gate compares the 28-sector benchmark at month-end with its trailing ten month-end average, including the current month. A value above the average retains the original ETF targets; other observations allocate to cash at the next trading-day open. Both variants share the gate, 3 bp fees, zero cash yield, and the same delisting conventions. The fixed ten-month rule adapts Faber's tactical-allocation approach.

For Q, the filter reduces annualized return by 8.28 pp and deepens maximum drawdown by 2.28 pp. For M, the corresponding changes are 8.67 pp and 7.77 pp. Average cash holdings are approximately 30.47% for the filtered portfolios. Volatility and fees decline while their realized drawdown paths worsen.

Table E1: Original and filtered strategy performance

Strategy	Ann. return	Max. DD	Volatility	Sharpe	Calmar
Q	24.62%	-44.76%	26.36%	0.967	0.550
Q+SMA	16.34%	-47.04%	23.14%	0.770	0.347
M	21.83%	-46.19%	25.92%	0.892	0.473
M+SMA	13.16%	-53.96%	23.21%	0.649	0.244

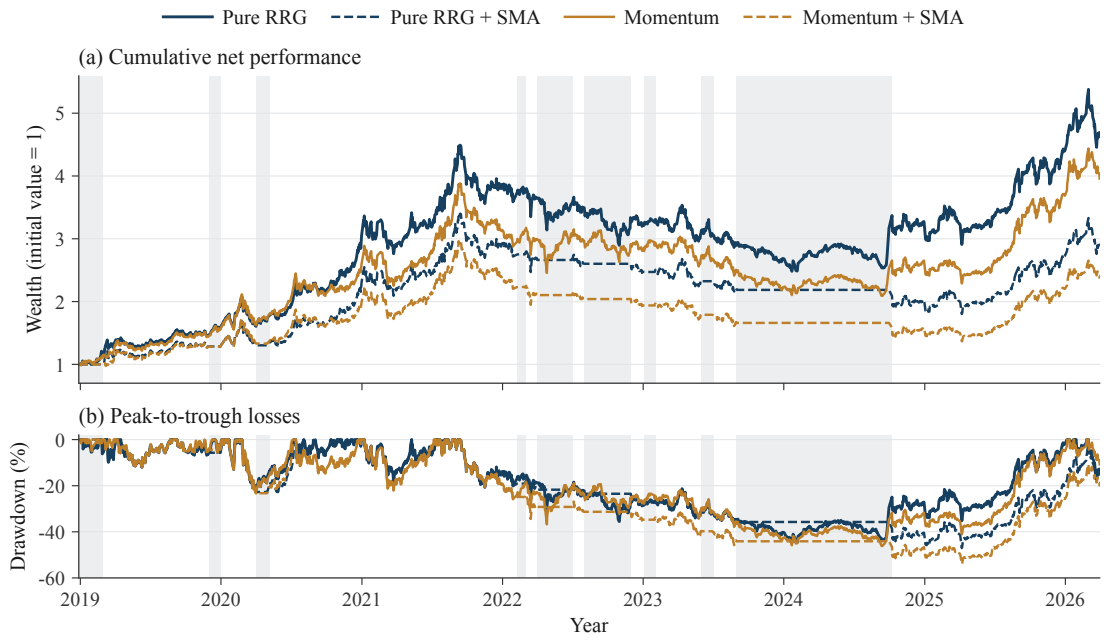


Figure E1: Trend-filter comparison. Solid lines denote the original strategies and dashed lines their ten-month moving-average variants. Shading denotes periods in which the common gate allocates the filtered portfolios to cash. Panel (b) reports each portfolio's drawdown relative to its own running maximum.

In September 2024, the closed gate misses Q's 17.58% return. Following re-entry on 8 October, filtered Q loses 9.78% that month, versus a positive 0.53% for the original. This episode illustrates the sensitivity of a lagged filter to rapid reversals.

Table E2: Dates of maximum drawdown

Strategy	Peak date	Trough date	Max. DD
Q	2021-09-13	2024-01-22	-44.76%
Q+SMA	2021-09-13	2025-04-07	-47.04%
M	2021-09-13	2024-09-11	-46.19%
M+SMA	2021-09-13	2025-04-07	-53.96%

The electronic supplement provides the complete monthly returns and gate states. Original NAV paths and execution checks reconcile with the archived results.

F Reproducibility index and supplementary records

The `tables/` directory accompanying this TeX project contains 19 full-precision CSV files. The tables below identify the complete underlying series as well as the test families. Detailed returns remain available electronically for replication and alternative summaries. The `figures/` directory contains English vector PDFs and corresponding 300-dpi PNG files. A common typographic and color scheme identifies strategies across all figures.

Table F1: Complete supplementary CSV inventory

File	Rows
`all_HAC_tests.csv`	18
`annual_returns.csv`	9
`benchmark_adjusted_regression.csv`	2
`cost_pairwise_tests.csv`	24
`cost_sensitivity.csv`	16
`grid_multiple_tests.csv`	54
`historical_membership_coverage.csv`	10
`main_performance.csv`	4
`month_dependency.csv`	4
`monthly_returns_complete.csv`	87
`paired_block_bootstrap.csv`	12
`parameter_grid.csv`	27
`primary_pairwise_tests.csv`	6
`rolling_12m_returns.csv`	76
`rolling_summary.csv`	4
`sma10_drawdown_periods.csv`	4
`sma10_market_filter.csv`	88
`sma10_monthly_comparison.csv`	88
`sma10_performance.csv`	4

Main source outputs: `results/rrg_tushare_repaired_20261107/`.

Core scripts in the project's `src/` directory:

- Data and execution: `rrg_repair_inputs.py`; `rrg_repair_backtest.py`.
- Robustness: `rrg_report_robustness.py`.
- Inference and risk control: `rrg_report_inference.py`; `rrg_sma10_overlay.py`.

Audit records cover settings (`run.json`), input repair (`input_audit.json`), execution (`validation.json`), checksums (`input_file_hashes.json`) and readback (`final_readback.json`).

Input database SHA256:

`e71a0043112ba173307eac16acbecc1f72c232bb1780ddac455d32c02a407524`

Fixed upstream commit:

`3f83a77f4c7996ed8185b8979a42c4372c1d811f`

The group's work comprises data integration and repair, extending the first-quadrant rule to ETF allocation, harmonizing implementation and benchmarking, and the additional empirical checks. Signal, exposure, optimization and core execution functions are reused from the pinned implementation.

References and methodological sources

Western Securities Research Center (2026). *Index-based Allocation Research Series (6): Sector Research under the Relative Rotation Graph Framework*. 25 May. Chinese report.

hugo2046. *QuantsPlaybook*. Fixed version above. `https://github.com/hugo2046/QuantsPlaybook`.

Tushare-compatible interface: authorized local data archive. Coverage and definitions are documented in Appendix A.

Faber, M. (2007; updated 2013). *A Quantitative Approach to Tactical Asset Allocation*. The Journal of Wealth Management. Source pointer retained in the existing experiment: `https://ssrn.com/abstract=962461`.

`statsmodels`: HAC-robust covariance and Holm multiple-comparison implementation, as recorded in the local inference scripts. `arch`: circular time-series block bootstrap implementation, as recorded in the local inference scripts.